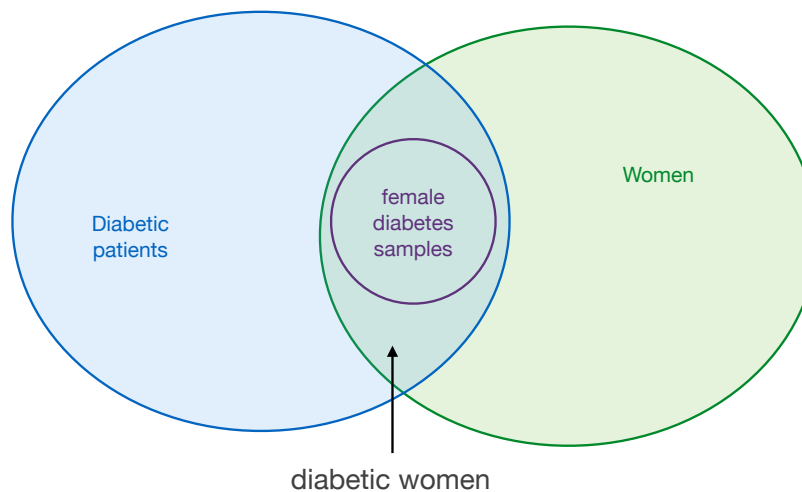


5. Distributions

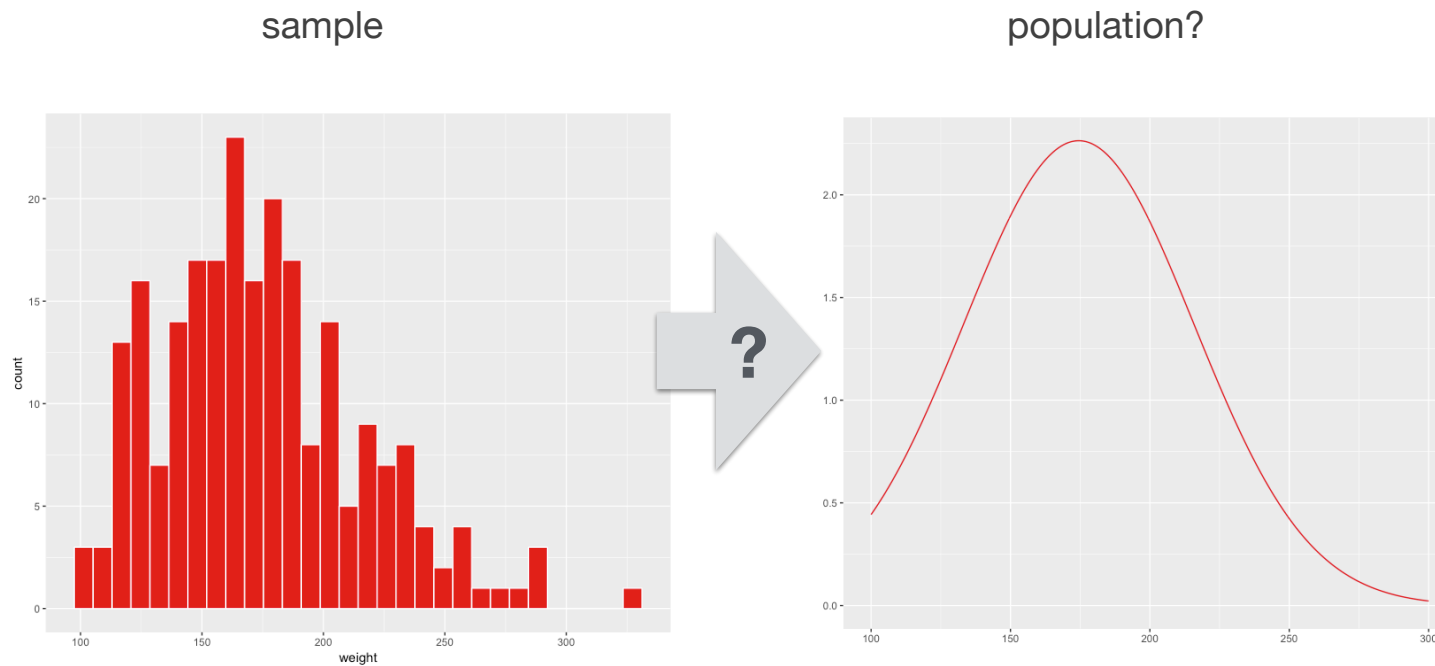
Principles of statistical inference

- so far, we have worked with datasets that represent **samples** of a more **general population**
 - sample: 403 diabetes patients → population: Diabetic Patients
 - sample: 152 ALL patients → population : All ALL patients
- We assume that this sample represents well the larger population
→ **representative sample**



sample is chosen based on the population of interest....

Diabetic patients



inference:

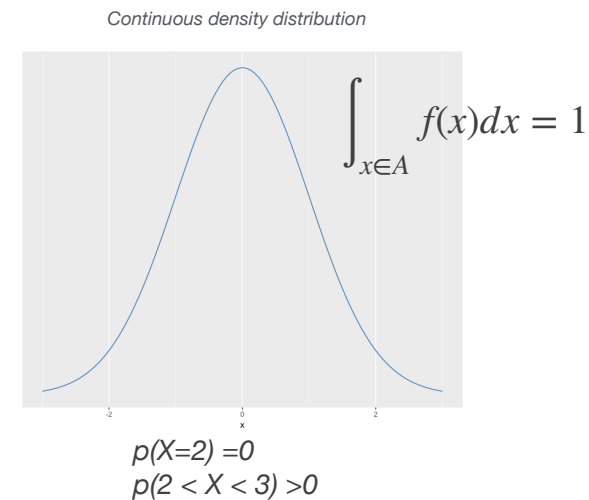
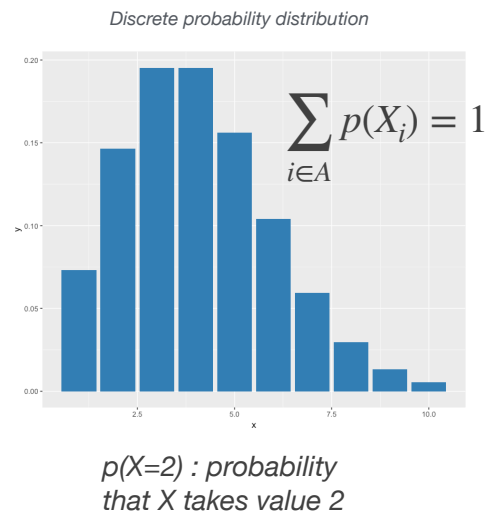
How can we estimate the parameters
(mean, spread,...) of the population based
on the sample?

Random variable

- Sample = vector of values ("**realizations**")
- Population = **random variable**, describing the (random) issue of an experiment or measurement
- Notations
 - $X, Y, \dots$ = **Random variable** (continuous or discrete)
example: values obtained by rolling a dice, electoral behavior in Germany, increase in survival after treatment
 - $x, y, \dots$ = **Realizations** (numerical or categorical)
example: values after three throws, votes of person A,B,C, increase in survival of specific patient cohort
- Random variables can be
 - **continuous** (weight, height, time,...)
 - **discrete** (counts)

Random variables

- Random variable X can be described using
 - **expectation $E(X)$** : “theoretical” mean of the entire population
 - **Variance $Var(X)$** : “theoretical” variance of the entire population
 - **Density distribution** (continuous RV) $f(X)$
 - **Probability distribution** (discrete RV) $p(X)$

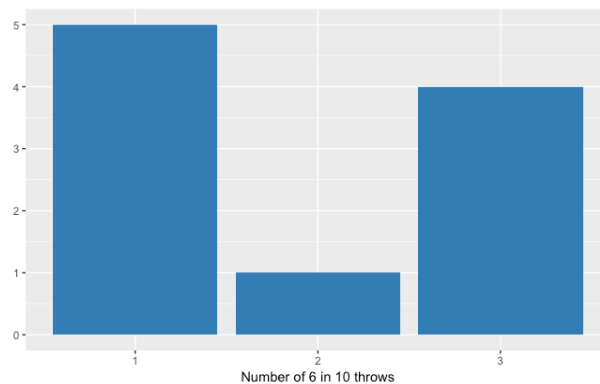


Example of a discrete RV

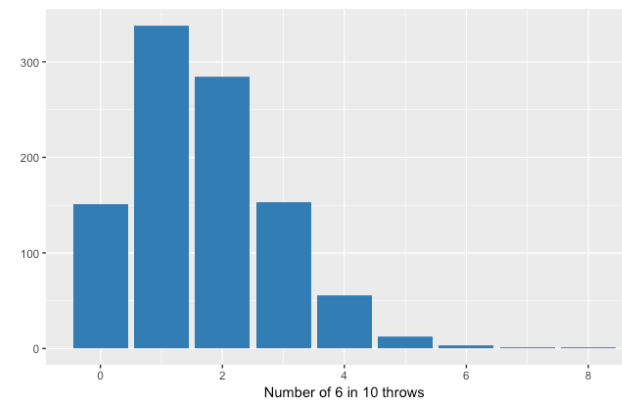
- Random variable X : number of 6 obtained in 10 dice throws

- $X \in \{0, \dots, 10\}$

10 games



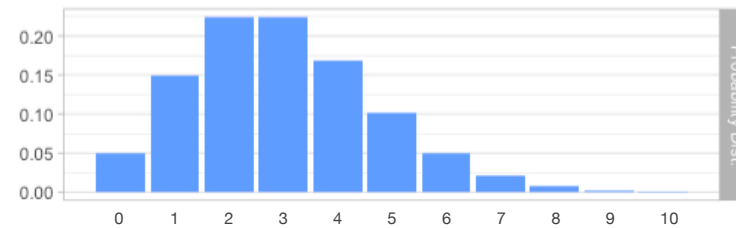
1000 games



$E(X)$? $\text{Var}(X)$?

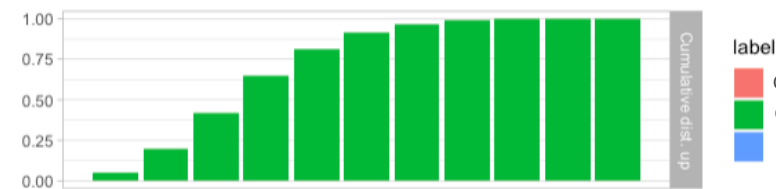
Cumulative distribution function (CDF)

- Example: **discrete RV**
- Probability distribution $P(k)$



- Cumulative distribution

$$F_X(k) = P(x \leq k)$$

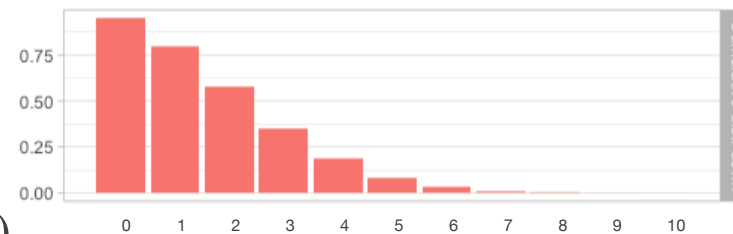


label

- Comp. cumul. distribution
- Cumulative distribution
- Probability distribution

- Complementary cumulative distribution

$$\bar{F}_X(k) = p(x > k) = 1 - F_X(k)$$

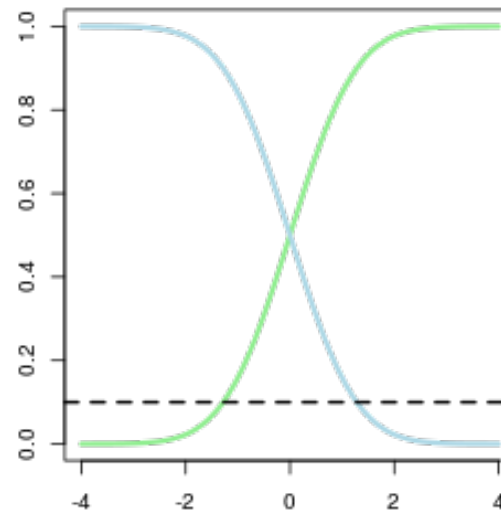
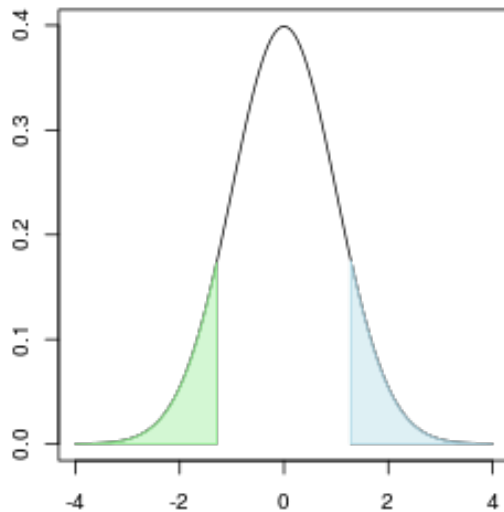


Cumulative distribution (CDF)

- **Continuous** random variable X
- density distribution $f(X)$
 - cumulative distribution
 - complementary cumulative distribution

$$F_X(x_0) = P(x \leq x_0)$$

$$\bar{F}_X(x_0) = P(x > x_0)$$



$$\bar{F}_X(x_0) = \int_{x_0}^{+\infty} f(x) dx$$

$$F_X(x_0) = \int_{-\infty}^{x_0} f(x) dx$$

Cumulative distribution of the uniform distribution?

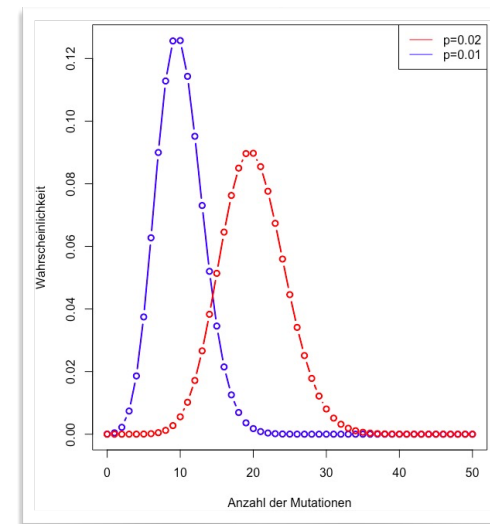
Binomial distribution

- Number of successes in L independent trials, each with probability of success p
- Example of DNA polymerase
 - error rate at each position p (constant, independent of the previous position)
 - sequence of length L
 - number of errors (= “successes”) ?

$$p(k) = \binom{L}{k} p^k (1-p)^{L-k}$$

$$E(X) = Lp$$

$$\text{Var}(X) = E(X)(1-p)$$



(how to compute the cumulative distribution?)

Poisson distribution

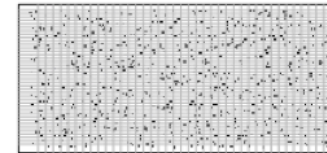
- Measures the number of events in a time period for a given **rate** of events
- Example: number of drops per tile during a rainshower $\lambda = 0.56$
- Rate needs to be constant and independent of previous period!

$$p(k) = \frac{\lambda^k}{k!} e^{-\lambda}$$

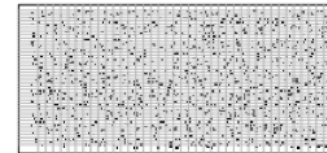
$$E(X) = \lambda$$

$$\text{Var}(X) = \lambda$$

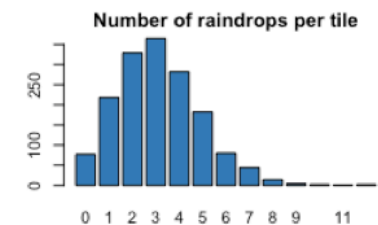
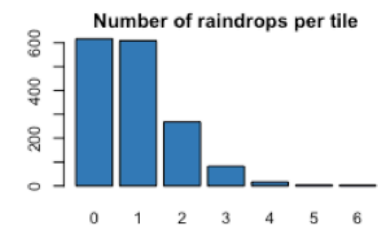
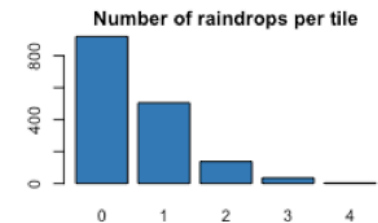
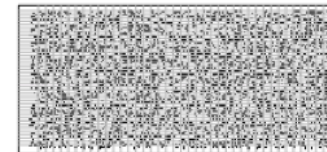
$\lambda = 0.56$



$\lambda = 0.94$



$\lambda = 3.125$



λ = drop per tile per unit time

Negative binomial

- Independent trials, constant probability p
- Different questions can be asked
 - ***Number of successes in L trials?***
→ **Binomial distribution** with 2 parameters (L, p)
 - ***Waiting time until r successes reached?***
→ **Negative binomial distribution** with 2 parameters (r, p)
 - Example: Taq-polymerase has accuracy $(1-p)$ with $p \sim 1/900$ (one error every 900 bases)
→ length distribution of sequences with $r = 3$ errors?

Negative binomial

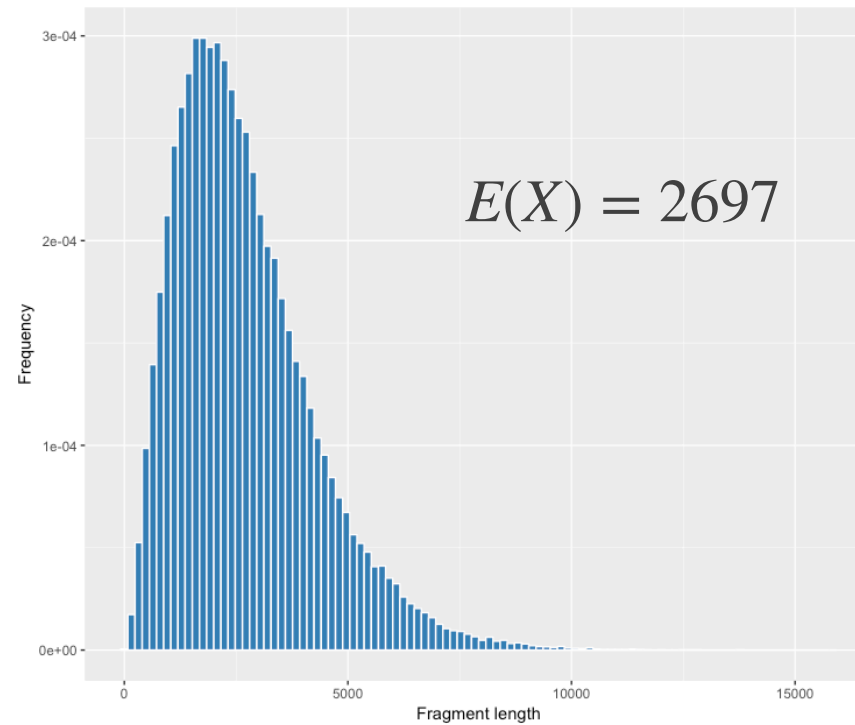
- $p = 1/900$; $r=3$ errors

$$p(k) = \binom{k+r-1}{r-1} p^r (1-p)^k$$

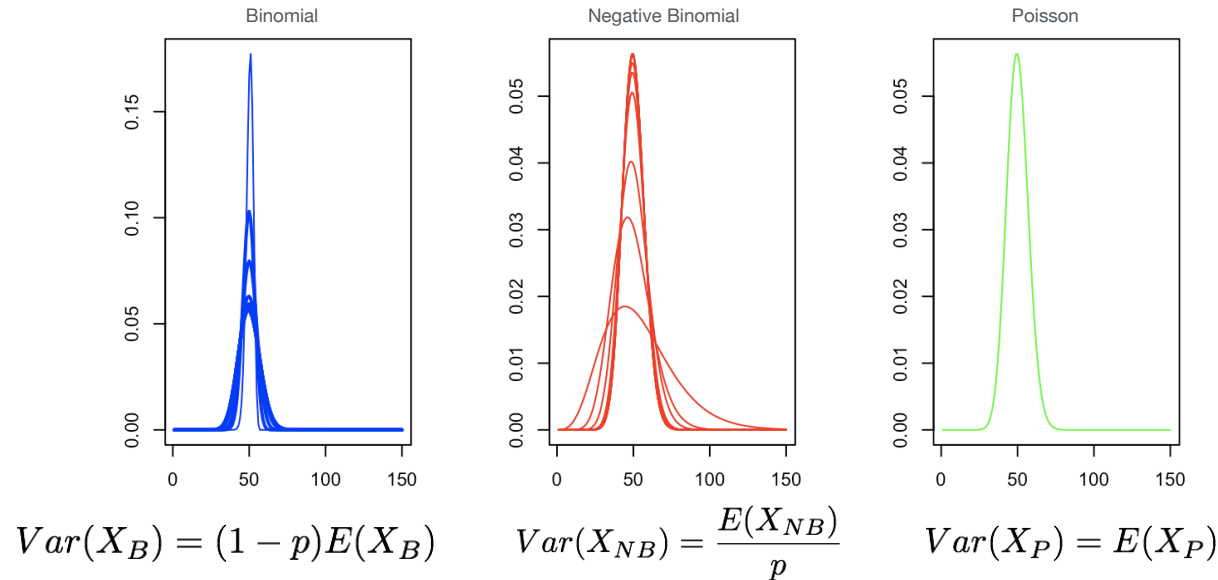
$$E(X) = \frac{(1-p)r}{p}$$

$$Var(X) = \frac{E(X)}{p} = \frac{(1-p)r}{p^2}$$

k = number of error-free bases



Negative binomial

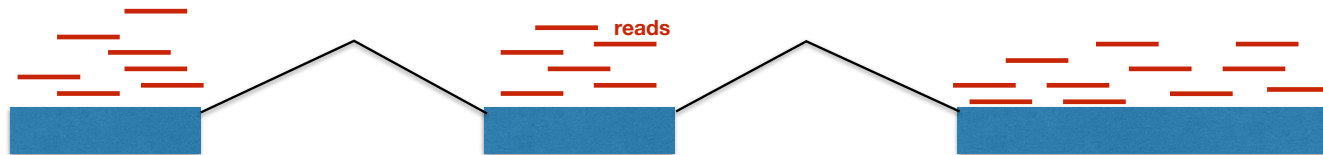


- The variance of the NB distribution can be made **arbitrarily wide**, by making p smaller

$$Var(X) = \frac{E(X)}{p} = \frac{(1 - p)r}{p^2}$$

- This can be useful in modelling processes showing an **over-dispersion**

Over-dispersion



- RNA-seq: mRNA molecules are fragmented and sequenced as short “reads” (e.g. 75-150 bp)
- reads are mapped onto the genome
- for each transcript, the number of reads is recorded

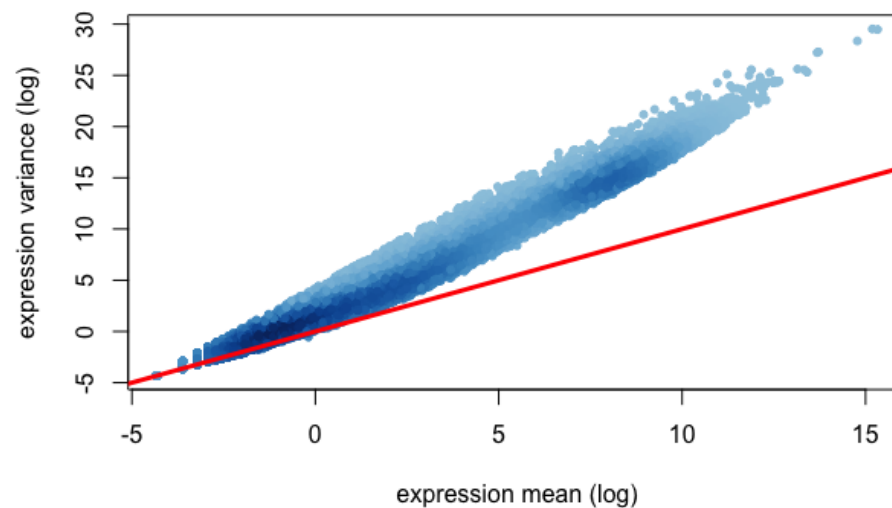
transcripts

samples / replicates

Description	GTEx-1117F-0226-SM-5GZZ7				GTEx-111CU-1826-SM-5GZYN				GTEx-111FC-0226-SM-5N9B8				GTEx-111VG-2326-SM-5N9I			
DDX11L1 3	4	1	1	0	2	1	3	5	1	0	1	3	1	1	3	
WASH7P 616	395	826	364	301	419	340	451	424	331	304	430	508	414	911	505	
MIR1302-11	2	1	1	0	1	0	2	3	1	1	1	0	0	1	0	
FAM138A 1	0	1	1	0	0	2	3	2	0	1	2	0	2	1	2	
OR4G4P 0	0	0	0	0	0	2	1	0	0	0	0	0	1	1	0	
OR4G11P 0	2	2	0	0	1	0	0	0	1	0	0	1	1	1	2	
OR4F5 0	0	0	0	2	0	0	0	1	1	0	0	2	1	1	0	
RP11-34P13.7	8	3	12	12	2	4	10	9	3	14	9	6	4	1	12	
CICP27 11	29	9	18	5	5	7	4	11	18	11	12	9	9	18	14	
AL627309.1	264	300	114	1364	227	276	518	52	91	105	558	354	91	195	587	
RP11-34P13.15	0	2	1	1	1	1	2	0	0	1	0	0	1	3	3	
RP11-34P13.14	5	0	0	16	4	9	9	0	0	2	16	10	1	8	12	
RP11-34P13.13	57	31	24	104	31	69	104	18	65	18	116	34	23	82	146	
RNU6-1100P	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	
RP11-34P13.9	1	0	0	1	1	0	0	0	0	1	1	0	0	0	0	
AP006222.2	52	113	168	7	100	46	25	29	44	256	35	152	79	159	511	
RP4-669L17.10	18	15	47	42	54	42	23	33	20	26	20	38	15	17	128	
RP4-669L17.8	1	1	2	2	6	0	3	0	3	0	0	0	1	1	2	
CICP7 5	6	5	2	2	1	2	4	4	4	2	7	11	8	3	13	
RP4-669L17.4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
OR4F29 0	1	0	0	0	3	0	1	0	0	0	0	0	2	1	2	
WBP1LP7 0	1	1	1	0	0	0	1	0	0	0	0	0	0	0	2	
AL732372.1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
<u>RP4-669L17.2</u>	0	0	0	1	1	0	0	2	1	1	0	1	1	0	0	
RP5-857K21.15	3	0	4	2	2	1	2	2	2	2	0	1	2	2	2	
RP4-669L17.1	3	3	1	0	2	2	1	0	5	2	2	6	1	2	5	

RNA-seq data

- each dot is a transcript; several replicates (~patients) available
- x-axis : mean number of reads per transcript (over all replicates)
- y-axis : variance of number of reads per transcript (over all replicates)
- **variance is larger than mean:** cannot be described by Poisson or binomial process!



if number of reads per transcript would be a Poisson distribution, dots should lie on the $y = x$ line

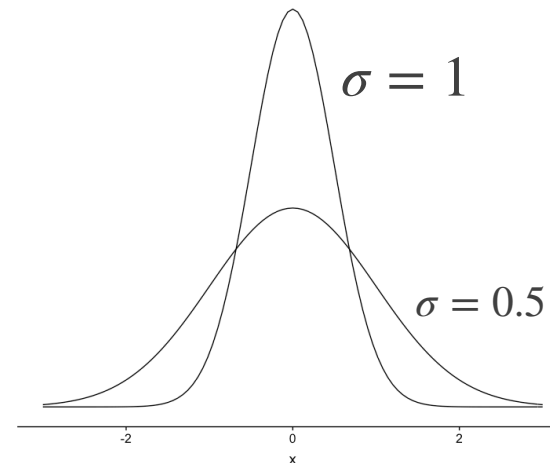
Normal distribution

- Continuous distribution
- plays a central role in statistics and data analysis due to *Central Limit Theorem*
- 2 parameters: expectation μ and standard deviation σ

$$f(X) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(X-\mu)^2}{2\sigma^2}}$$

$$E(X) = \mu$$

$$\text{Var}(X) = \sigma^2$$

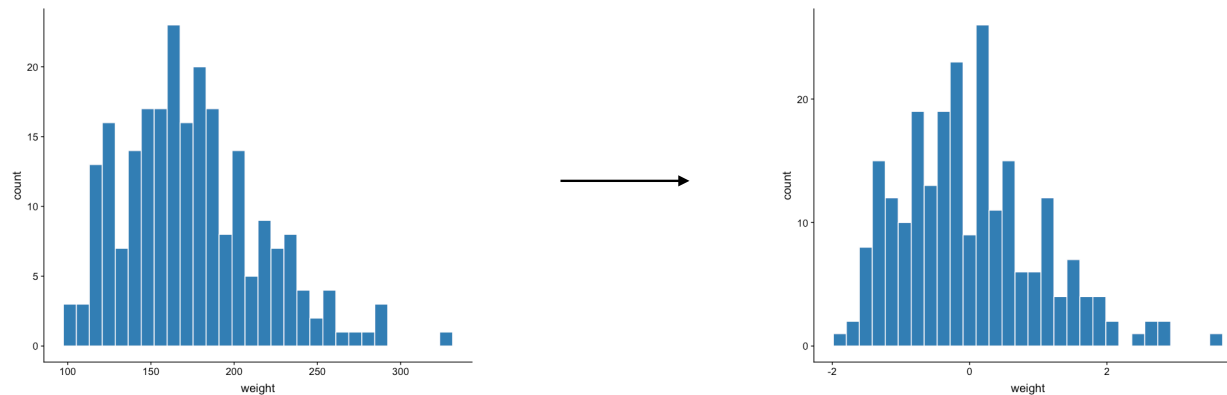


Normal distribution

- The normal distribution with $\mu=0$ and $\sigma=1$ is called the **Standard Normal distribution** $\mathcal{N}(0,1)$
- Every normal distribution $\mathcal{N}(\mu, \sigma)$ can be transformed into the SND through a Z-transformation

$$X \longrightarrow Z = \frac{X - \mu}{\sigma}$$

- By definition, the new RV Z has expectation 0 and variance 1
- The Z-transformation can be applied to any distribution!



Properties of SND

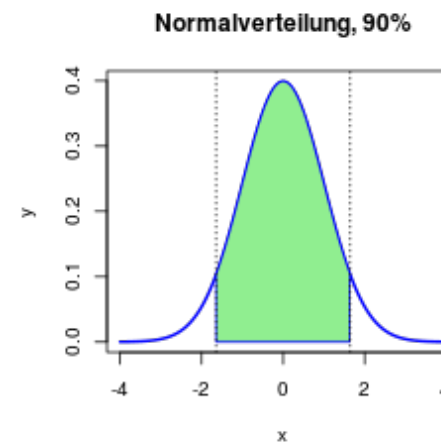
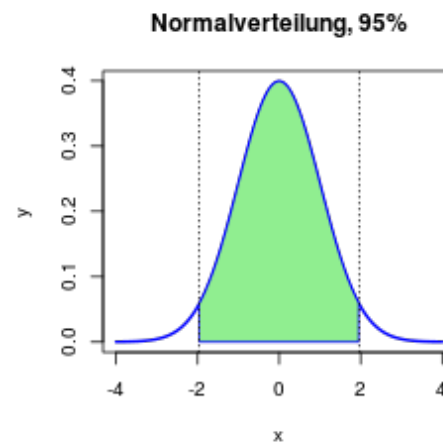
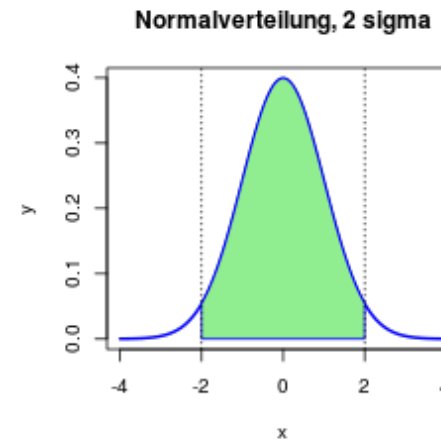
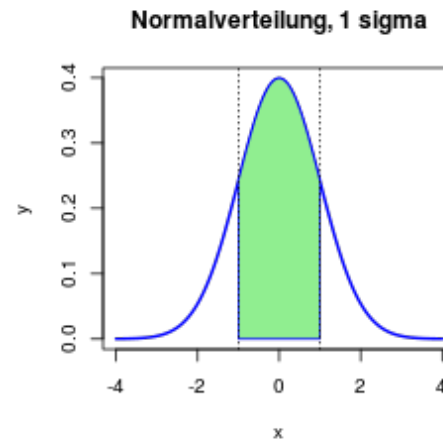
- the area represents...

- $[-1, +1] = 68\%$
- $[-2, +2] = 95.4\%$

- $[-1.96, +1.96] = 95\%$
- $[-1.64, +1.64] = 90\%$

- Critical values

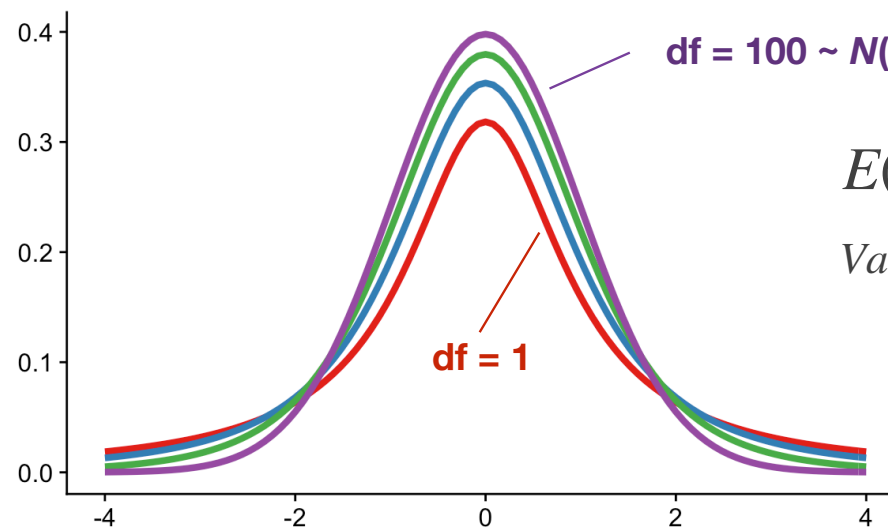
- $t_{95} = 1.96$
- $t_{90} = 1.64$



$$\int_{-\infty}^{+\infty} f(X) dX = 1$$

t-distribution

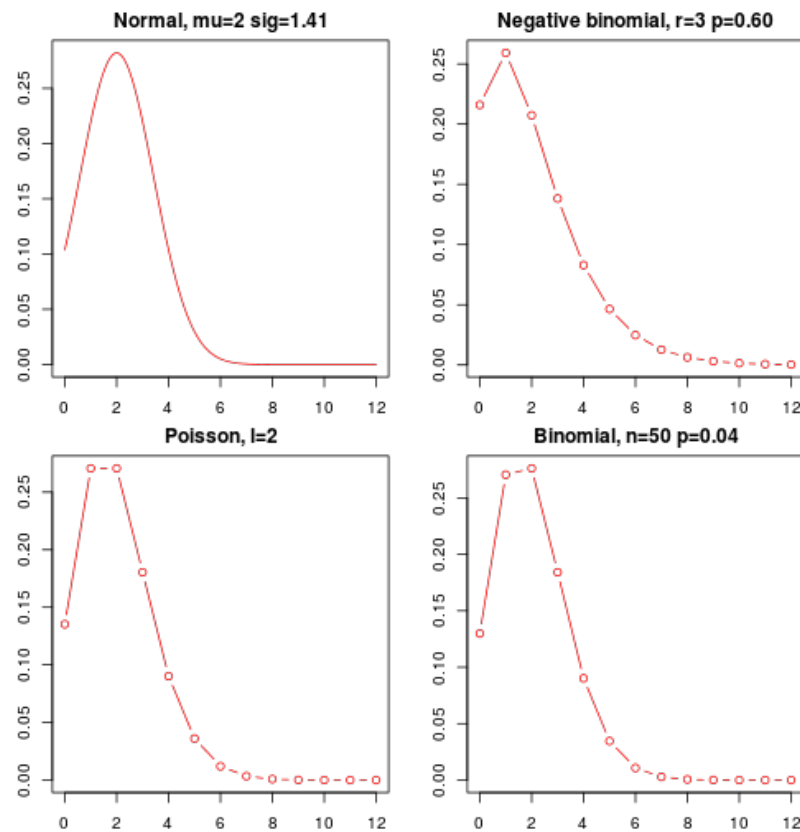
- Student's t-distribution is a continuous distribution describes the **distribution of mean values over small samples**
- Parameter: *degree of freedom (df)*
- Tends to the SND for large number of degrees of freedom



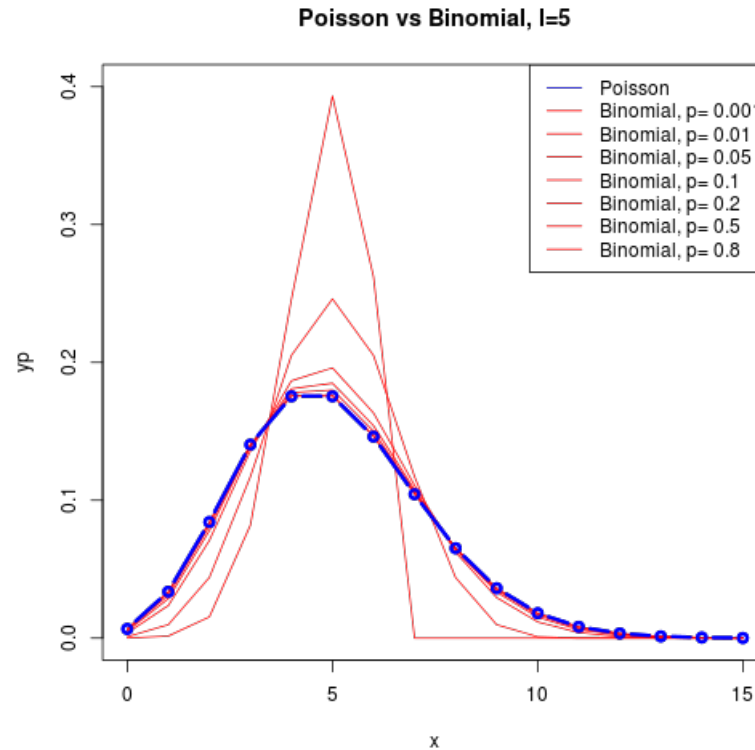
$$E(X) = 0$$
$$Var(X) = \frac{df}{df - 2} \quad \text{for } df > 2$$

Distributions are like an italian village....

- all are somehow related...

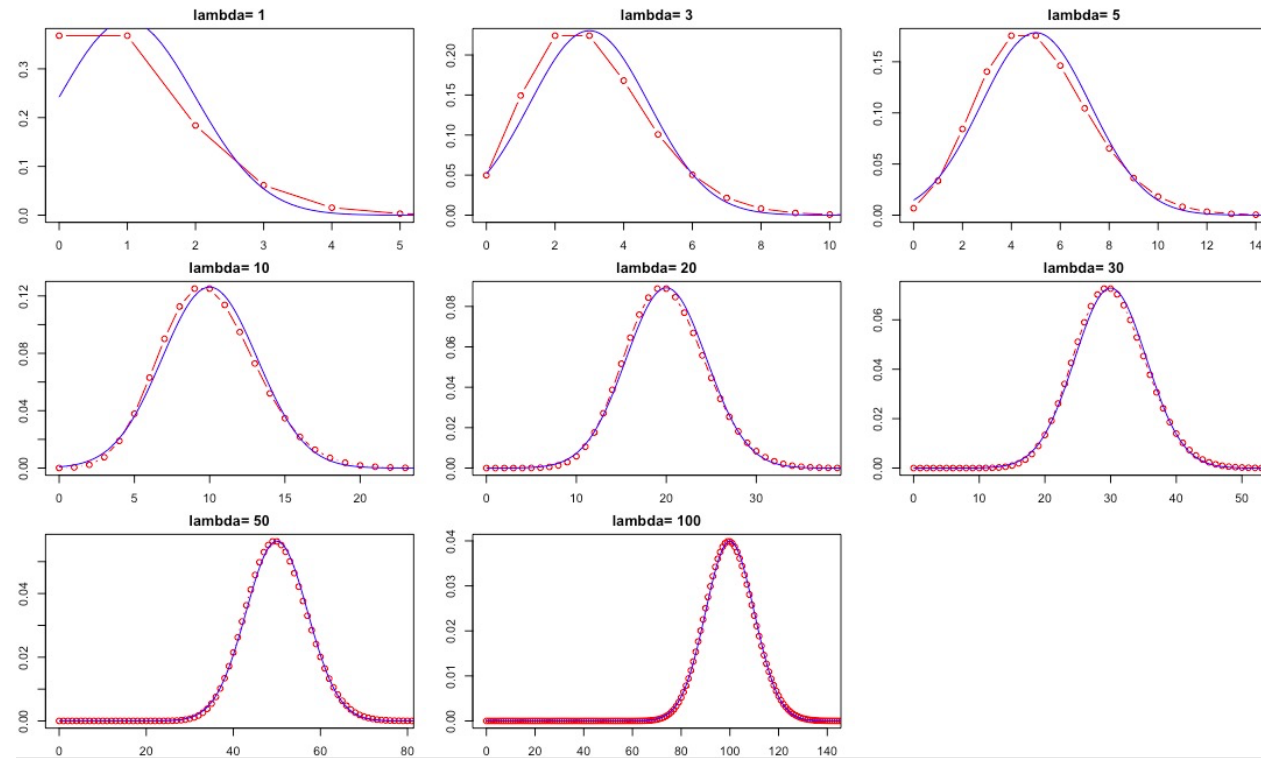


Poisson vs. binomial



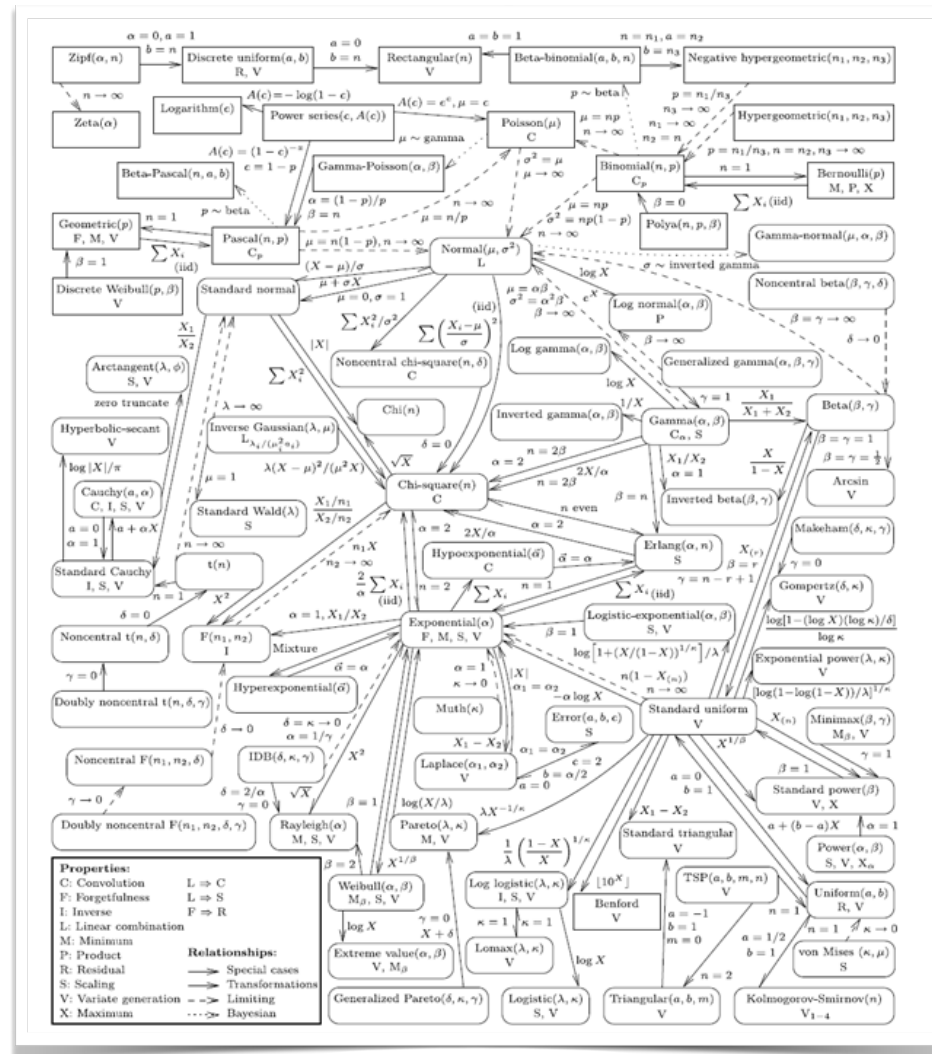
$$\text{Binom}(n, p) \longrightarrow \text{Pois}(\lambda = np)$$
$$n \rightarrow \infty; np = \lambda$$

Poisson vs. normal



$$Pois(\lambda) \xrightarrow[\lambda \gg 1]{} \mathcal{N}(\lambda, \sqrt{\lambda})$$

Distribution = italian wedding party!



Week 4 - stop and think

*First think about the questions on a sheet of paper;
then use your favorite AI to generate a small R code to illustrate these questions!*

1. You observe RNA-seq count data where $\text{Var}(X) = 50 \cdot E(X)$. If you consider binomial and negative binomial...
 - a. The data cannot follow any of these distributions
 - b. Only Negative Binomial with $p \approx 0.02$ could explain this over-dispersion
 - c. A mixture of Poisson distributions with different λ values could also produce this pattern
 - d. Both B and C are plausible explanations
2. Why does the t-distribution have heavier tails than the normal distribution for small sample sizes?
3. What exactly is preserved and what is lost in the Z-transformation? Think about: distribution shape, outliers, interpretability, and the relationship between multiple variables.
4. As λ increases, the Poisson distribution approaches a normal distribution. At what value of λ would you say they are "close enough" for practical purposes? Design a small simulation to test this.